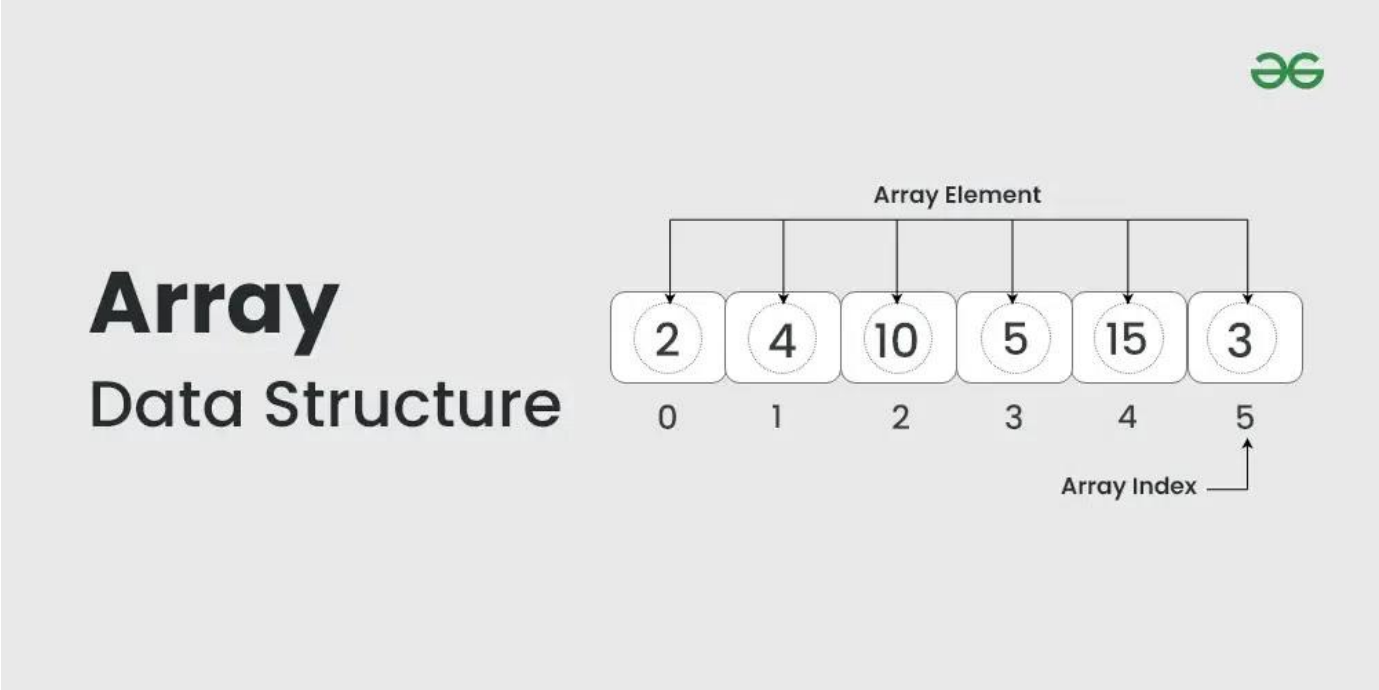
Vector Databases

What, Why & How?

Agenda

- What is a vector
- Embeddings
- What is a vector DB
- How does it work
- Why do we need a vector DB
- Relational DBs vs Vector DBs
- Types of vector DBs

What is a vector?



What is a vector?

Abstract representation of “something”

What is a vector?

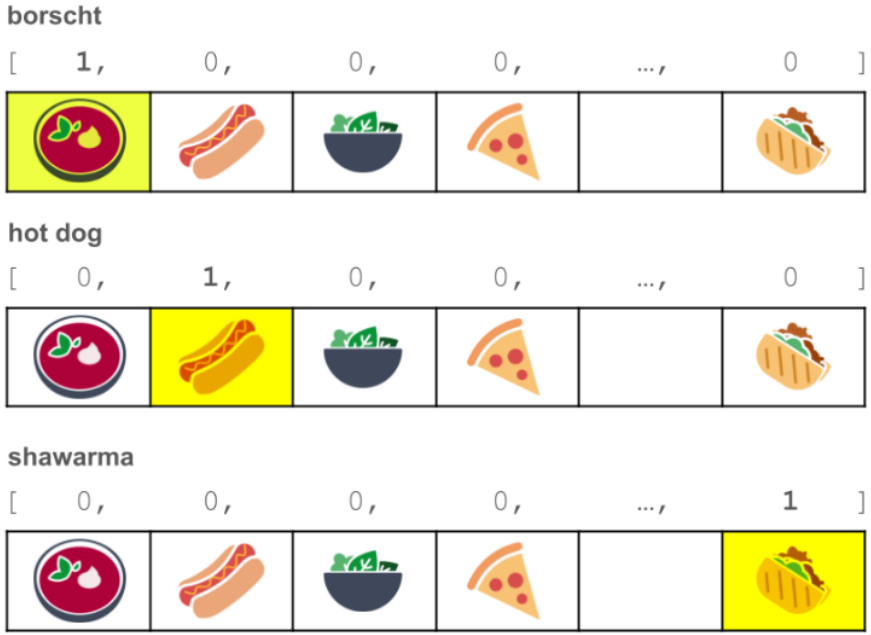
Abstract representation of “something”

Weekly gym schedule

Mo	Tu	We	Th	Fr	Sa	Su
1	0	1	0	0	1	0

What is a vector?

Abstract representation of “something”

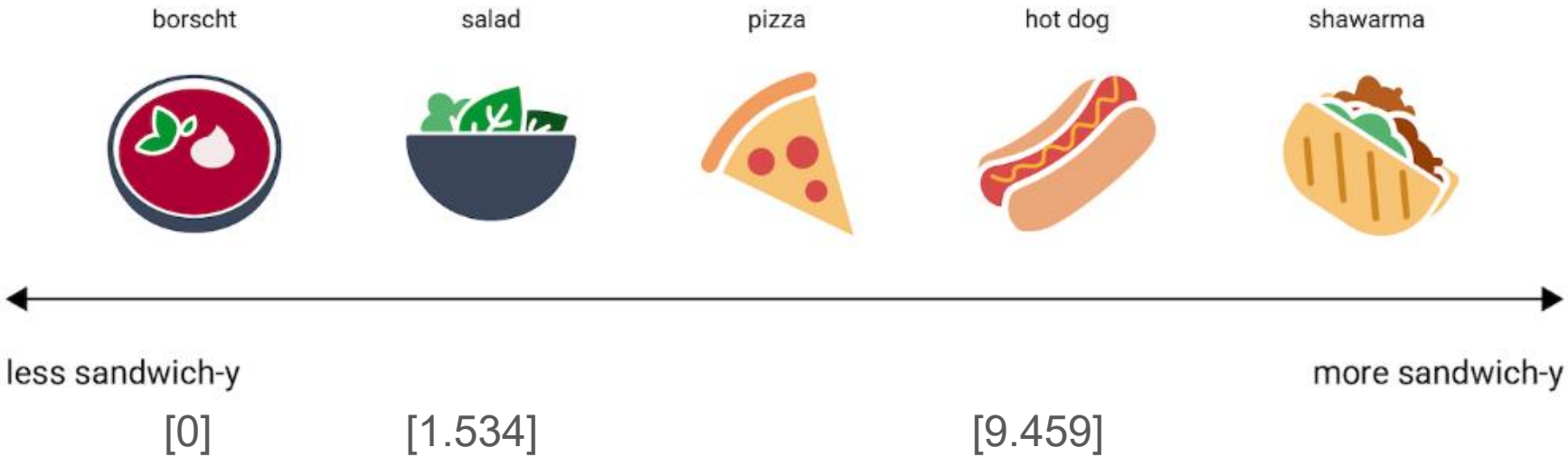


One-hot encoding

What is a vector?

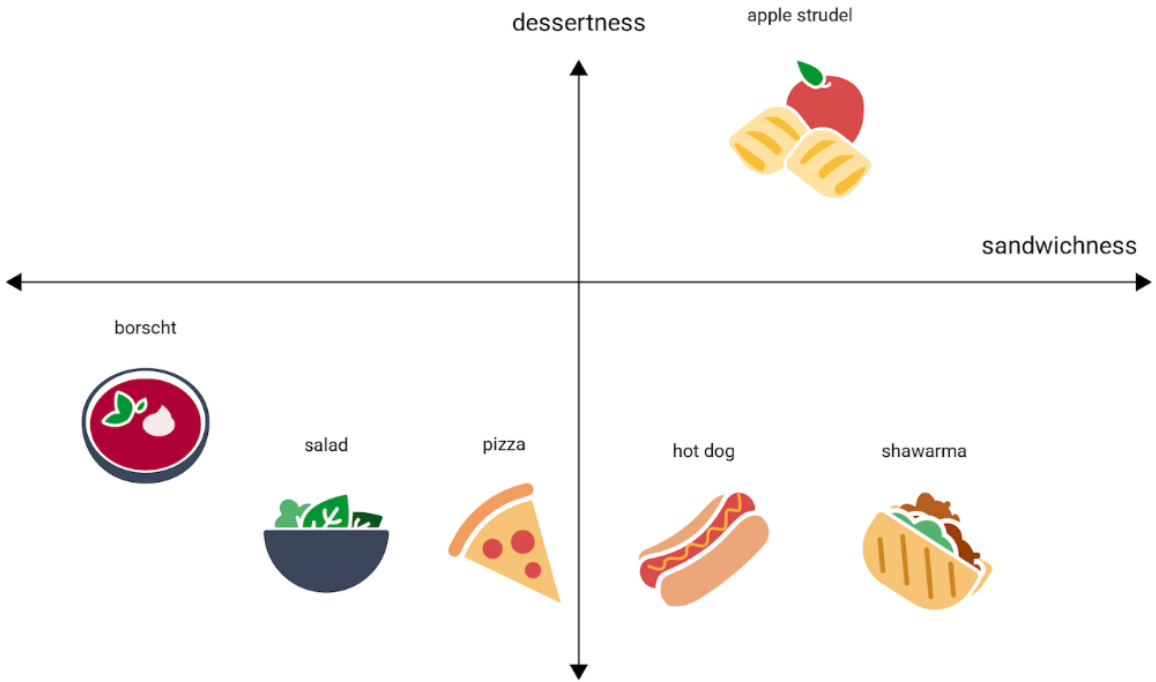
Abstract representation of “something”

One dimensional



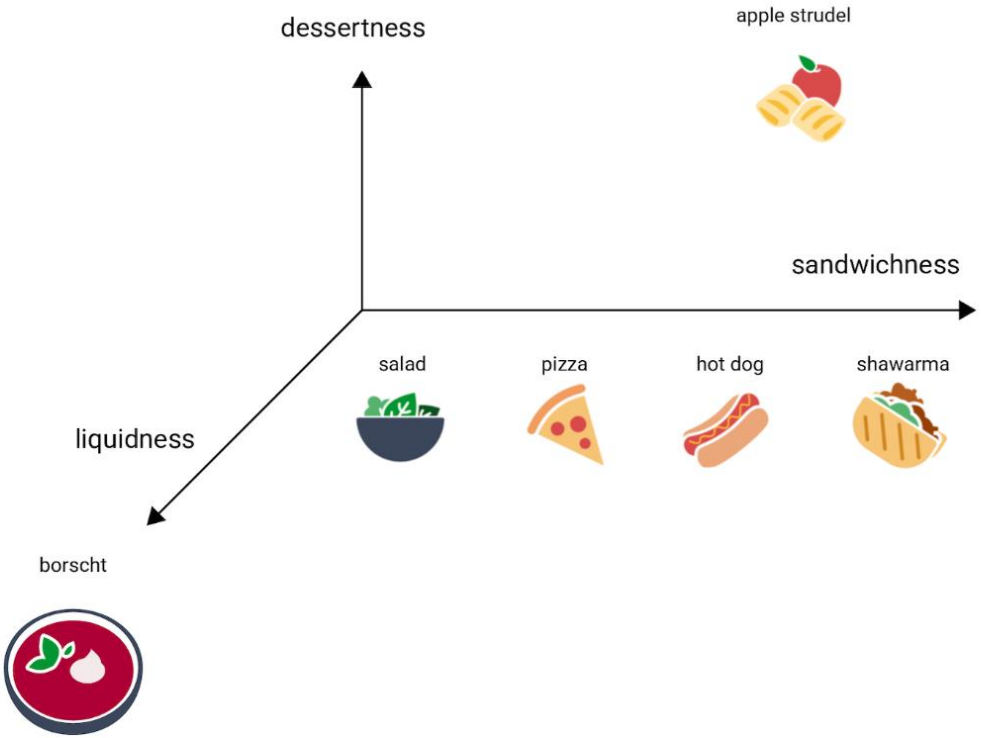
What is a vector?

Abstract representation of “something”



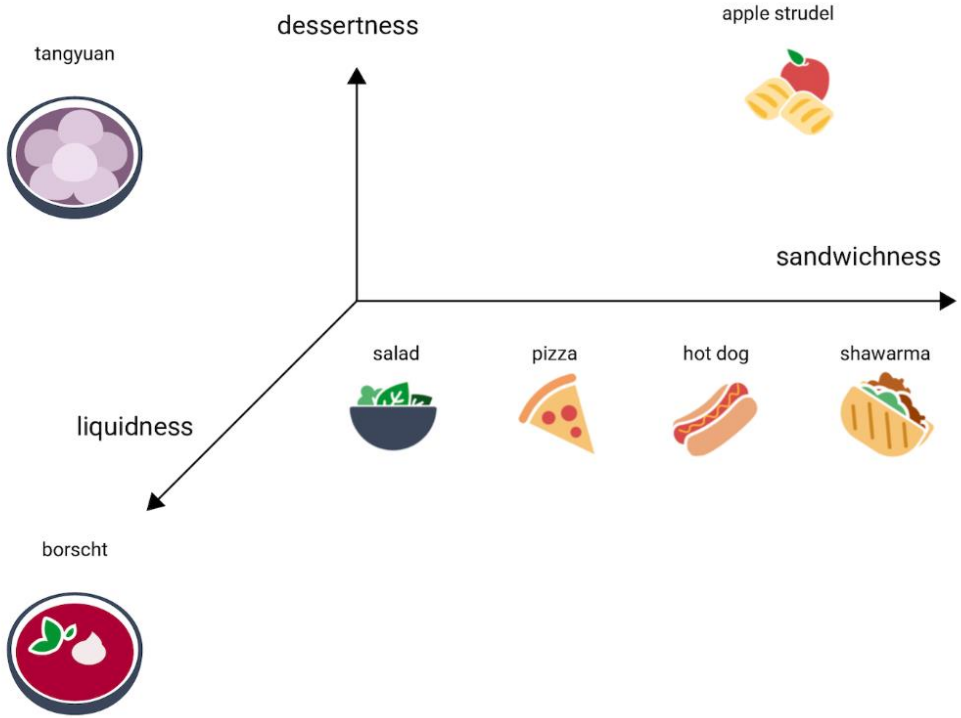
What is a vector?

Abstract representation of “something”



What is a vector?

Abstract representation of “something”



What is a vector?

Abstract representation of “something”

Weekly gym schedule

Mo	Tu	We	Th	Fr	Sa	Su
1	0	1	0	0	1	0

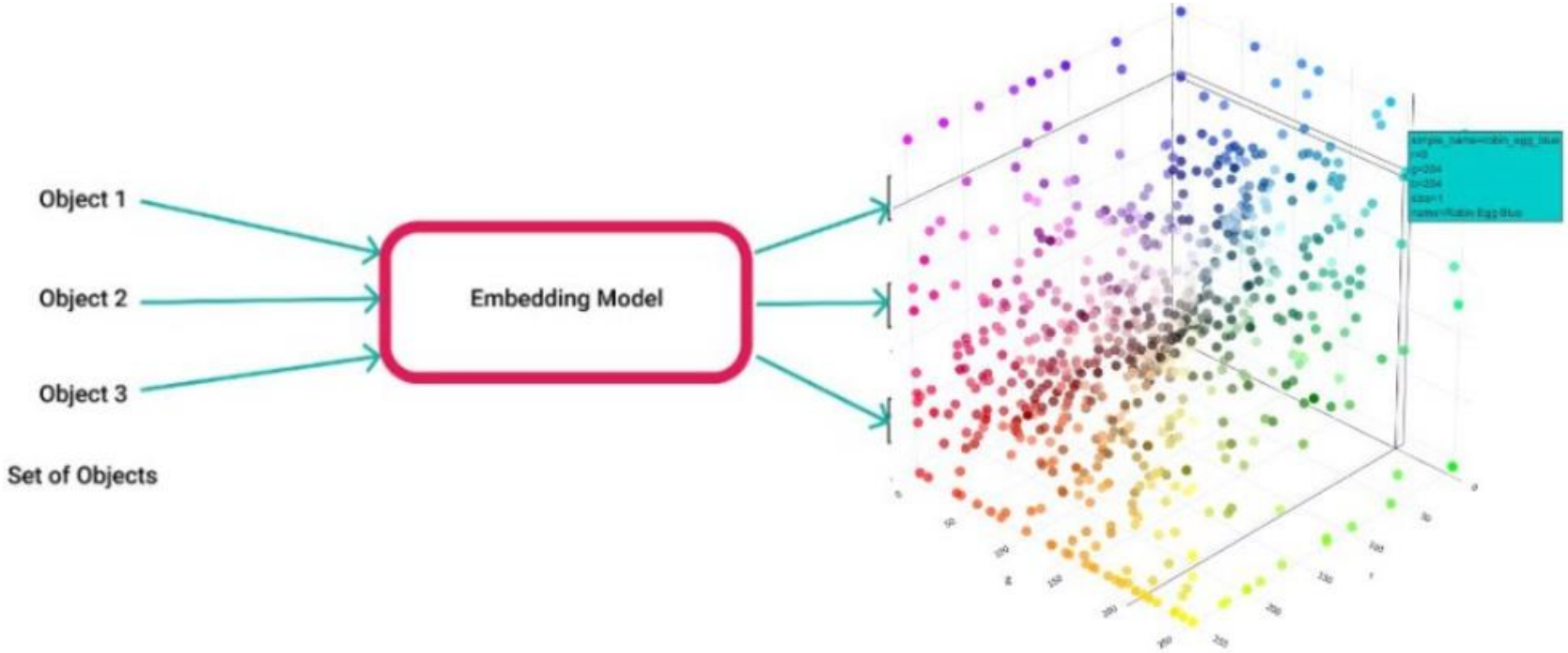
Embeddings

Vectorized representation of “something”

1	0	1	0	0	1	0
1	0	1	0	1	1	1
0	1	0	0	1	0	0
0	0	0	0	0	0	0

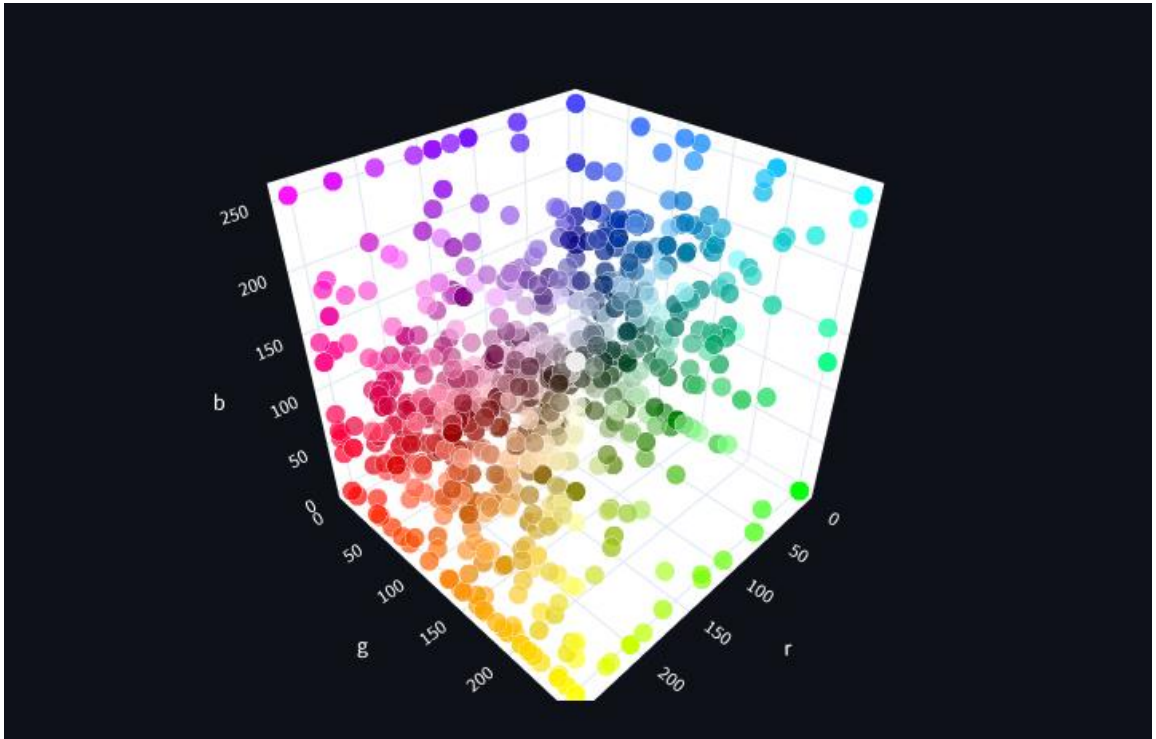
Embedding Models

An algorithm that generates vector representations of “things”



Embedding Models

Data represented in the Embedding Space

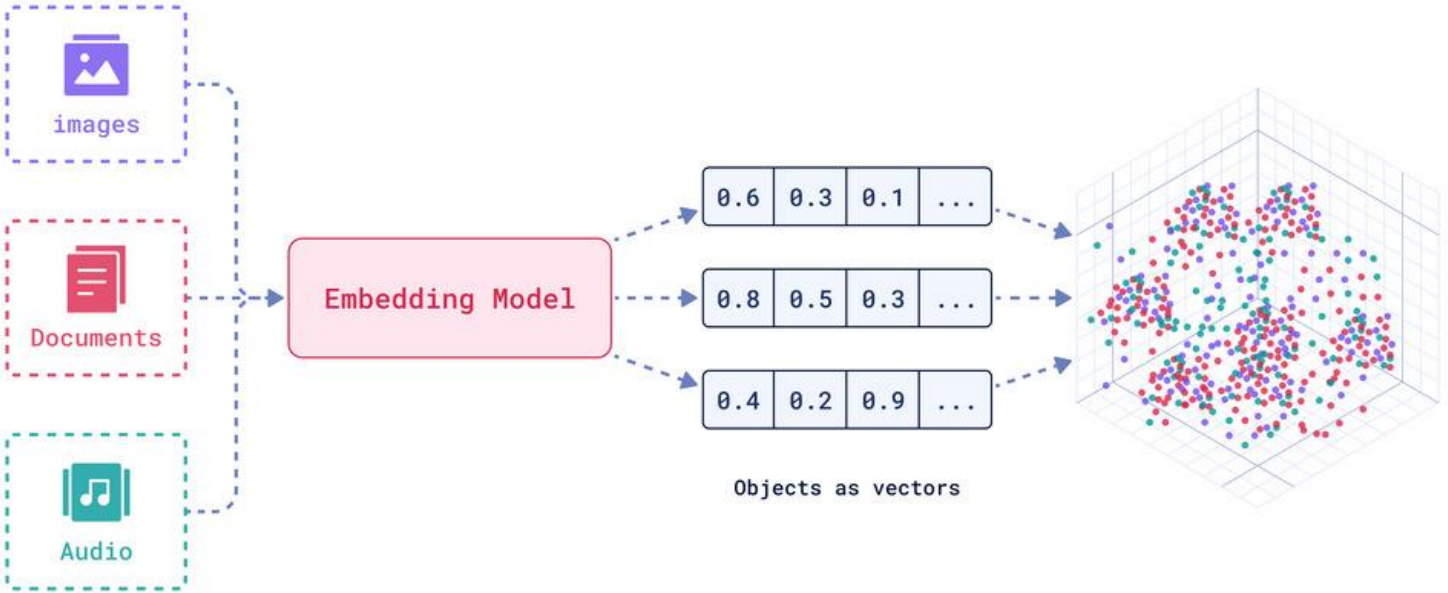


Embedding Models

What things can be embedded?

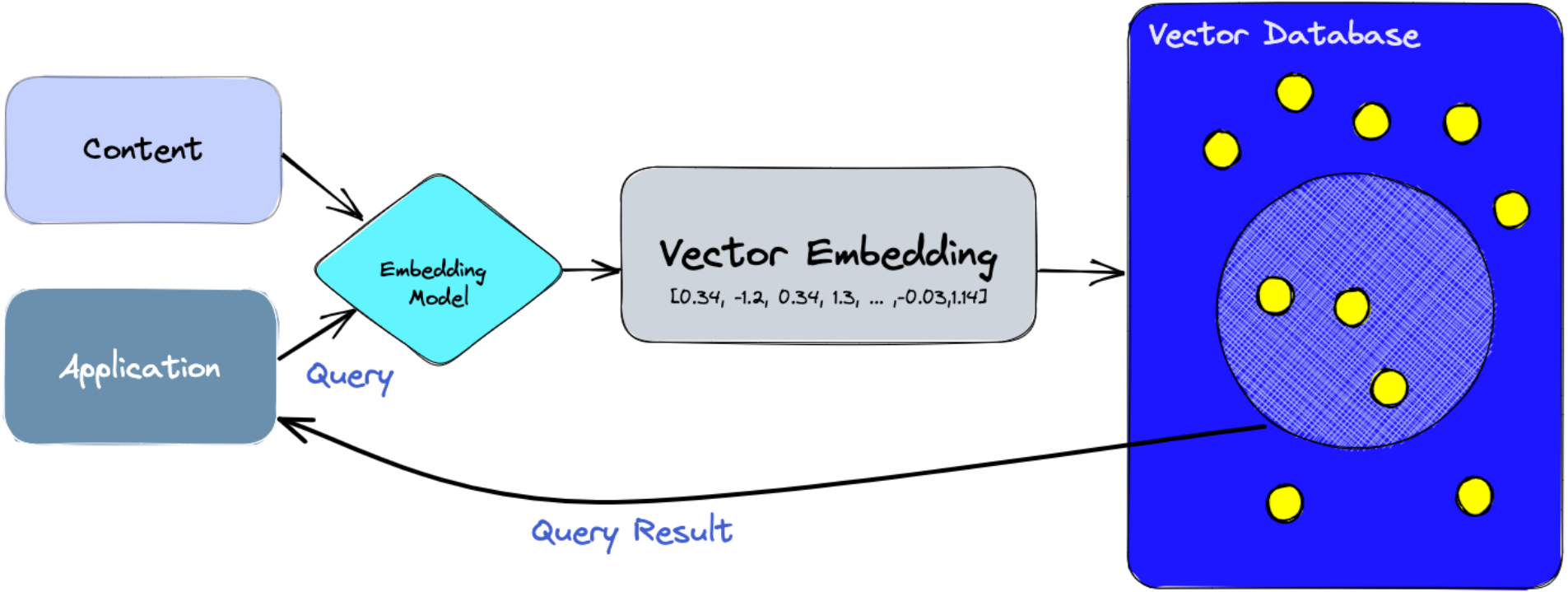
Embedding Models

What things can be embedded? EVERYTHING!



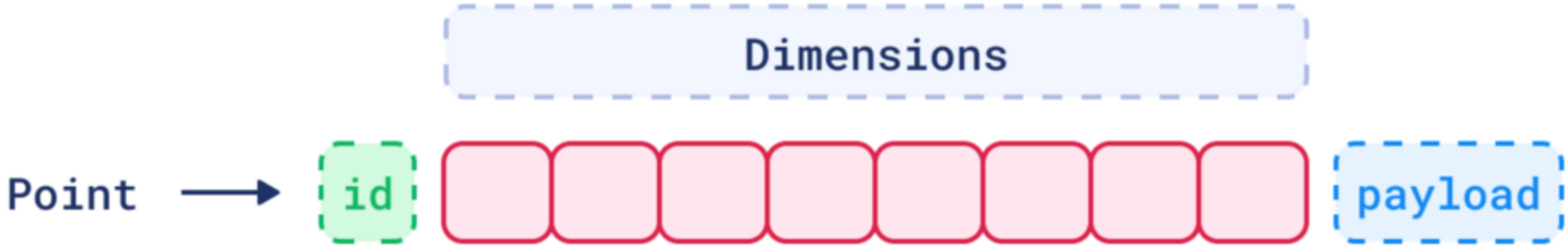
Vector Databases

A database specifically designed to work with vectors(embeddings)



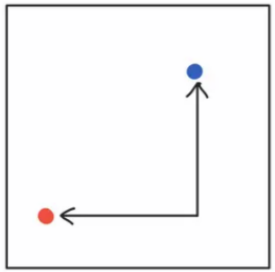
Data representation

Point = Id + Data itself + vector representation + metadata

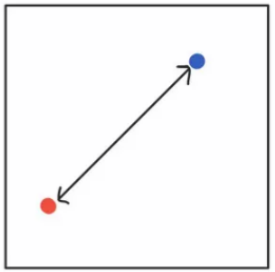


Distance metrics

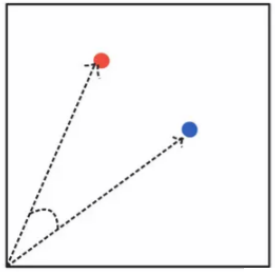
Relations between points in a space



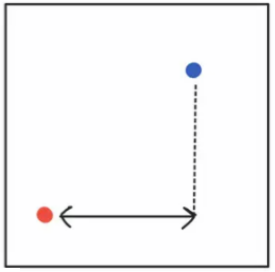
Manhattan



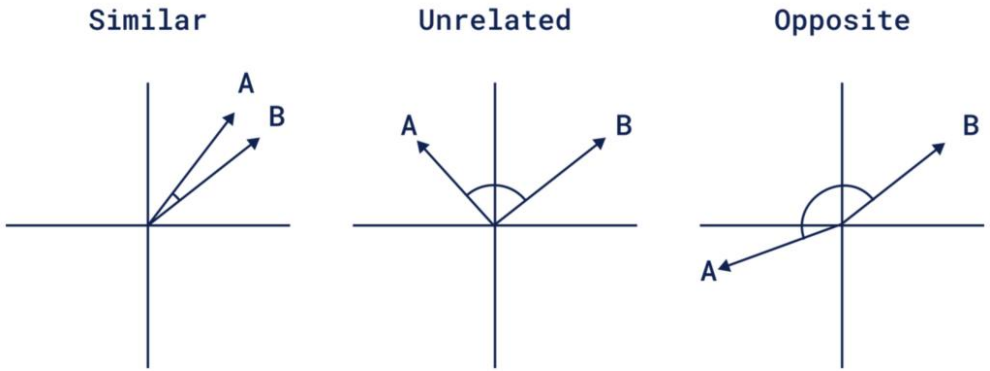
Euclidean

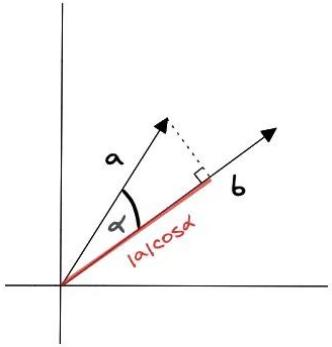


Cosine



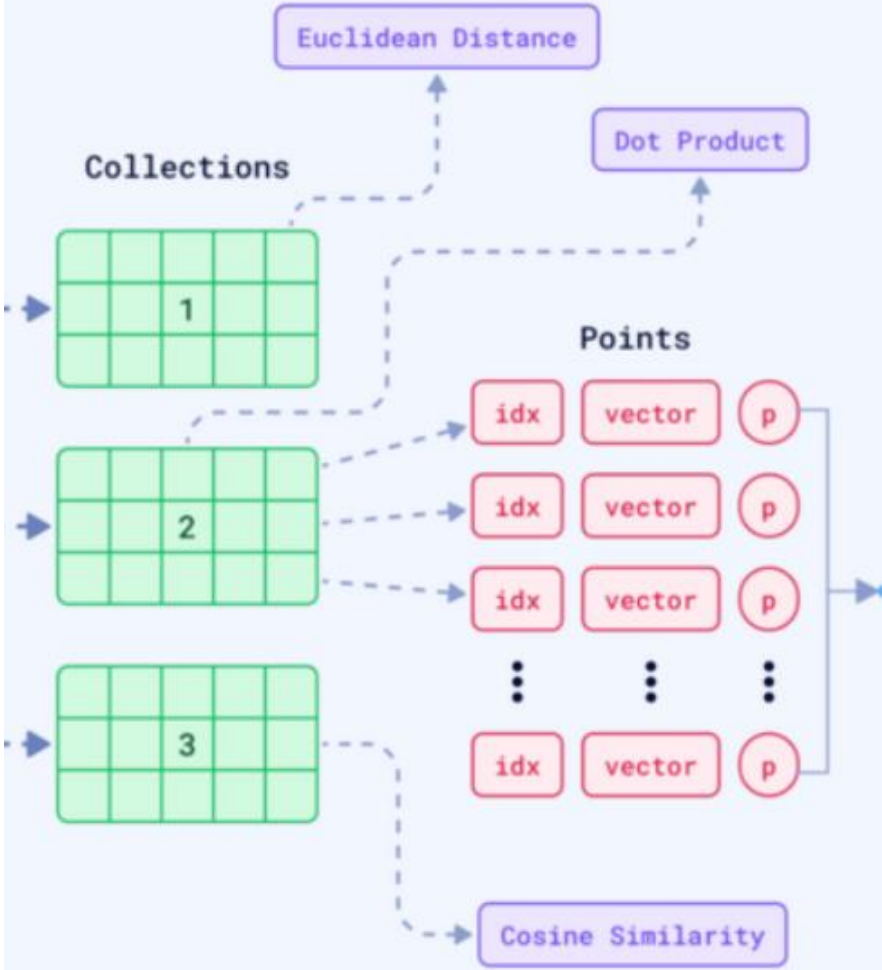
Chebyshev





Data storage

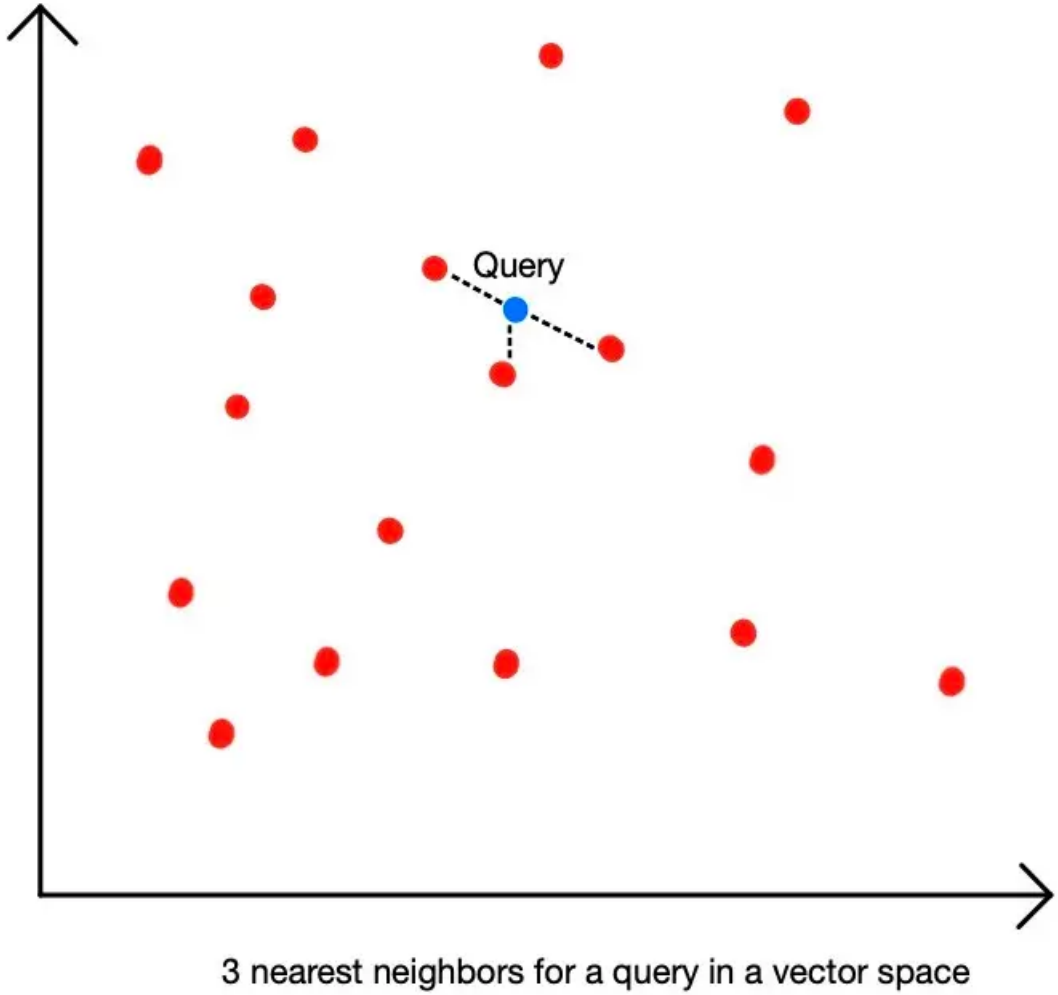
Collections = vector indexes



Indexing algorithms

Enabling fast querying with fast traversable data structures

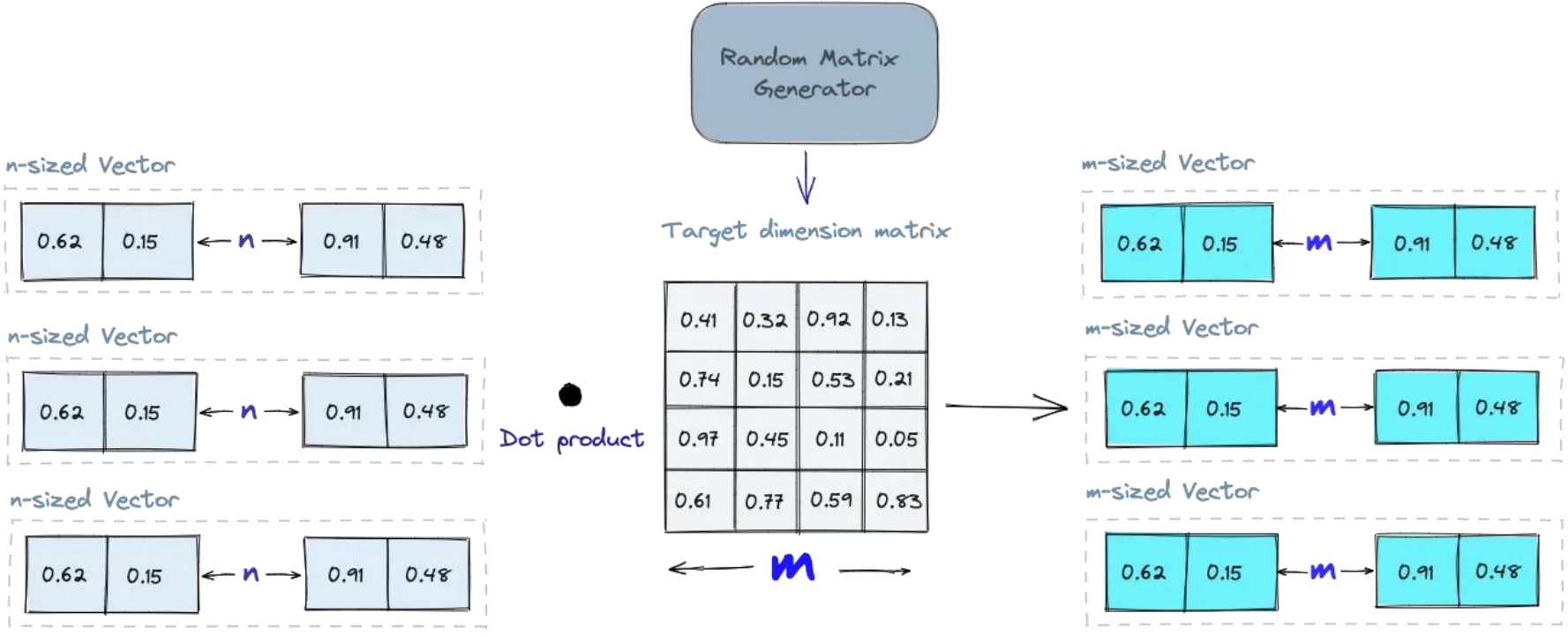
Approximate Nearest Neighbour(ANN)



3 nearest neighbors for a query in a vector space

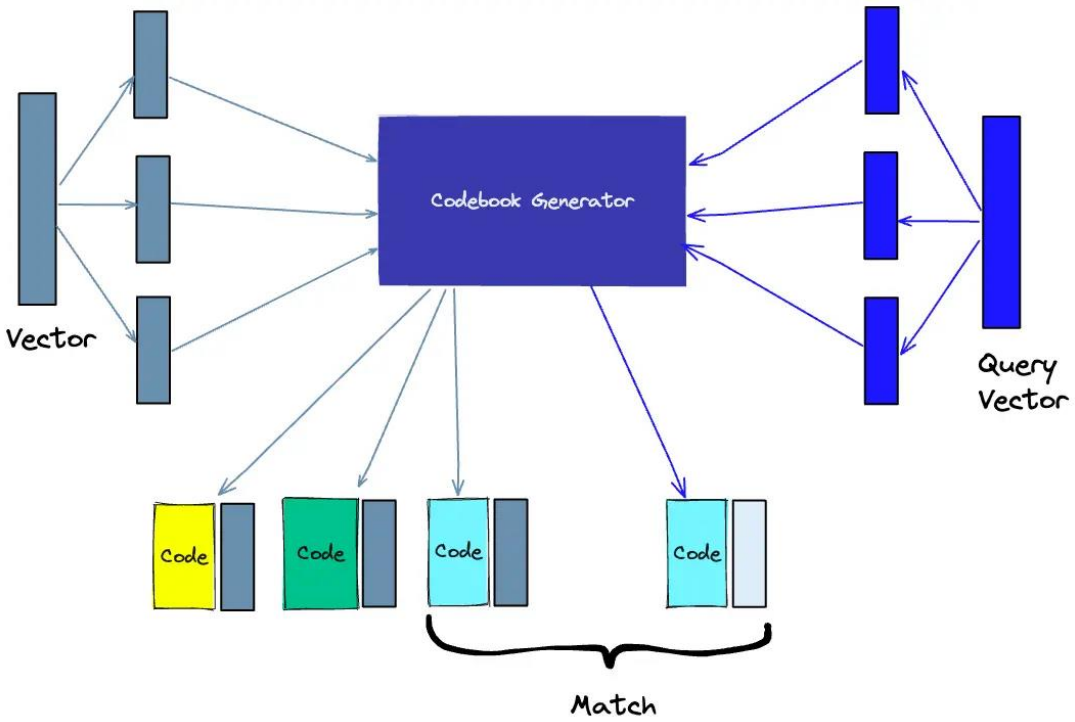
Indexing algorithms

Random Projection - dimensionality reduction



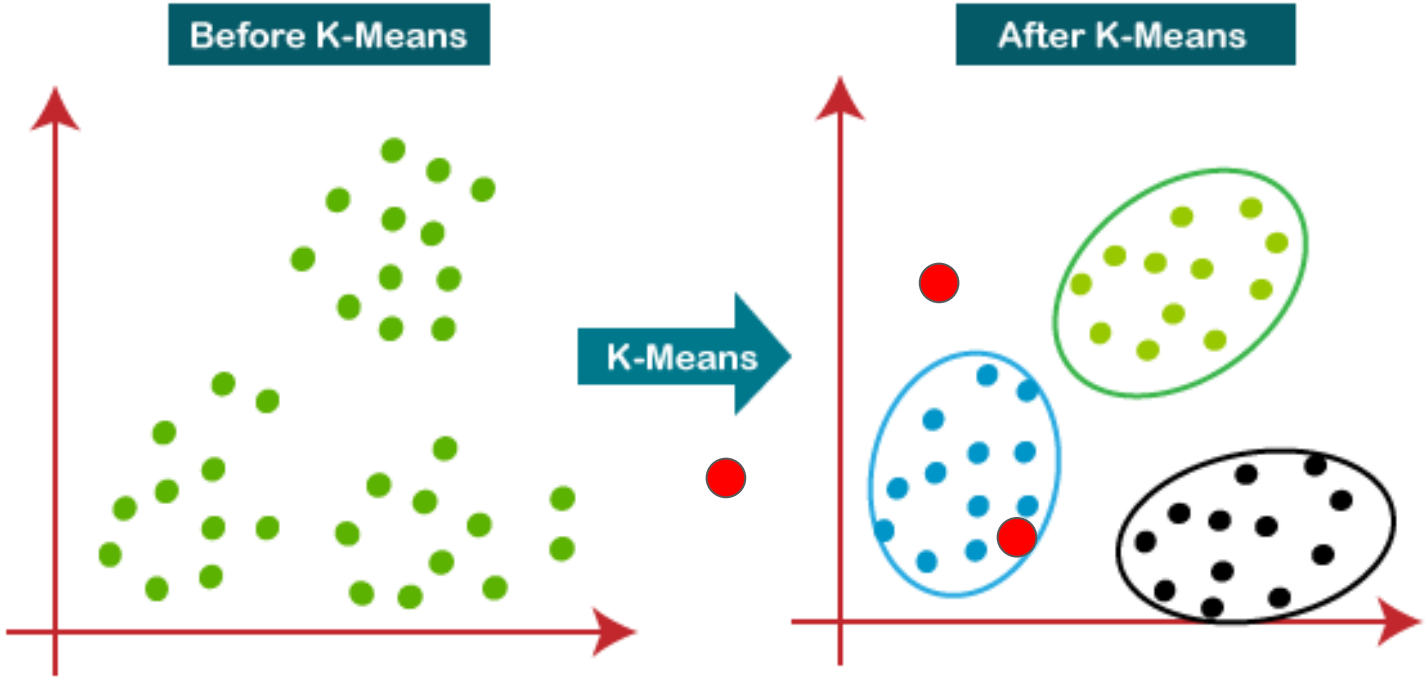
Indexing algorithms

Product Quantization - fractional hashing with k-means clustering for hashes = dimensionality reduction



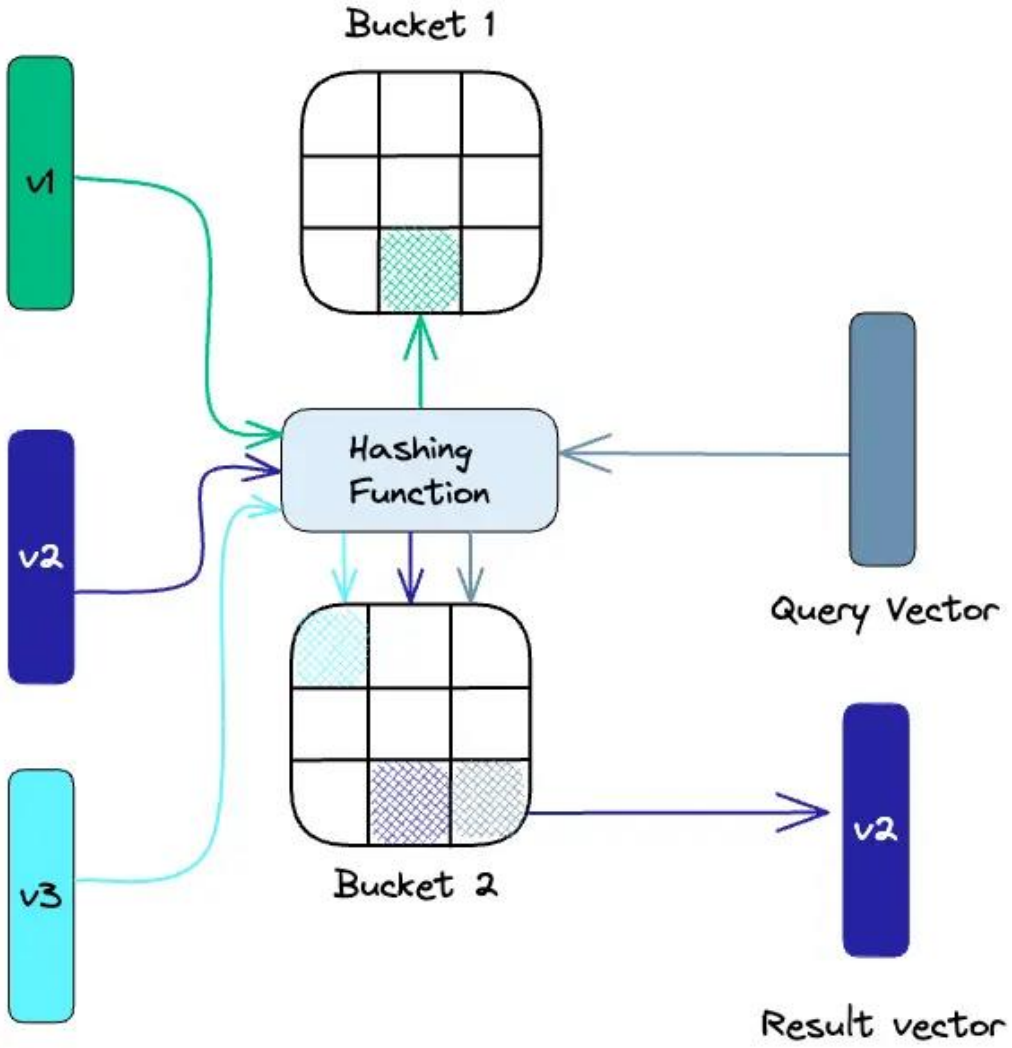
Indexing algorithms

Product Quantization - fractional hashing with k-means clustering for hashes = dimensionality reduction



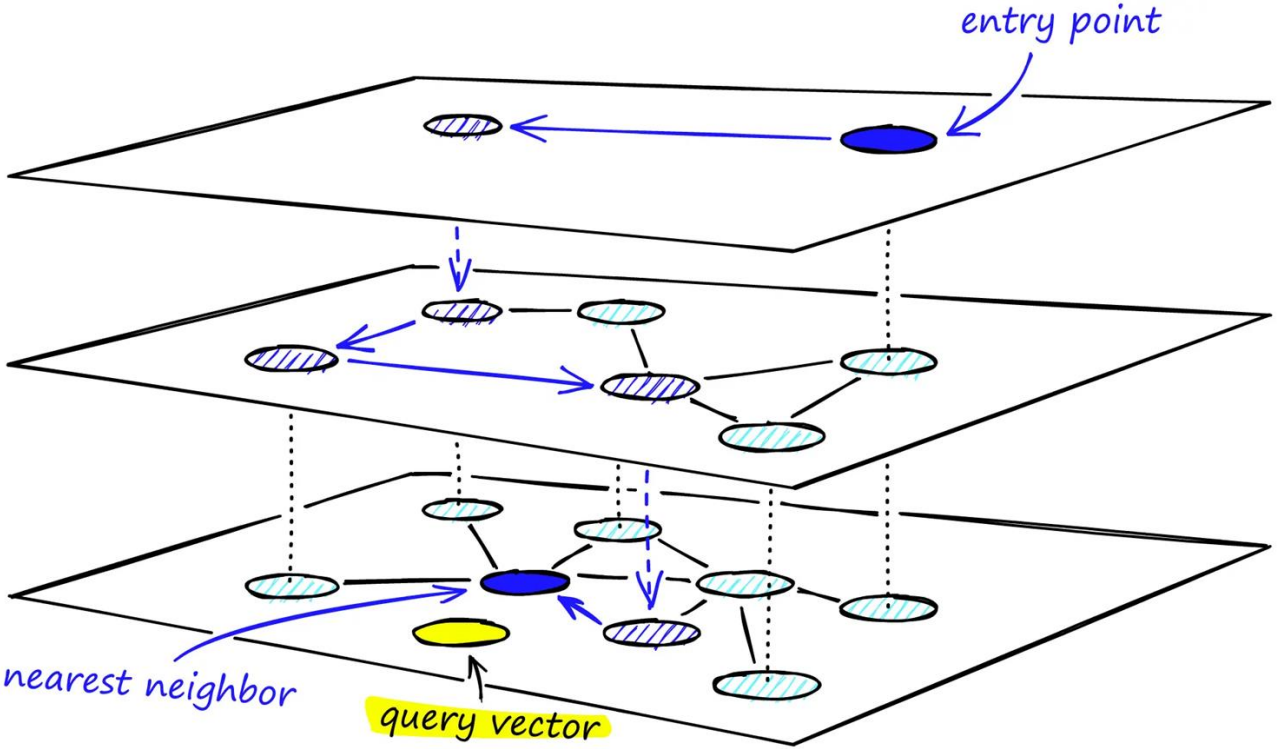
Indexing algorithms

Locality-sensitive hashing - reducing the search space



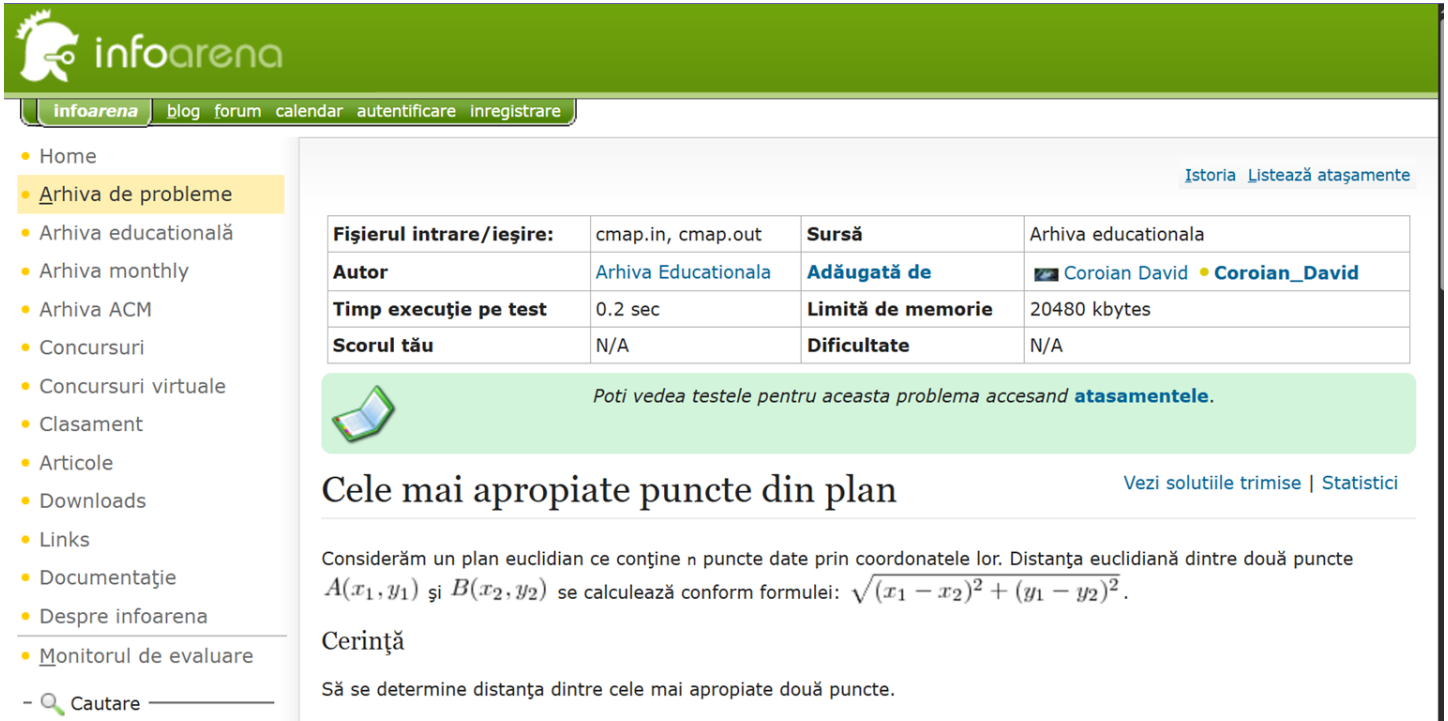
Indexing algorithms

Hierarchical Navigable Small World (HNSW) - state of the art



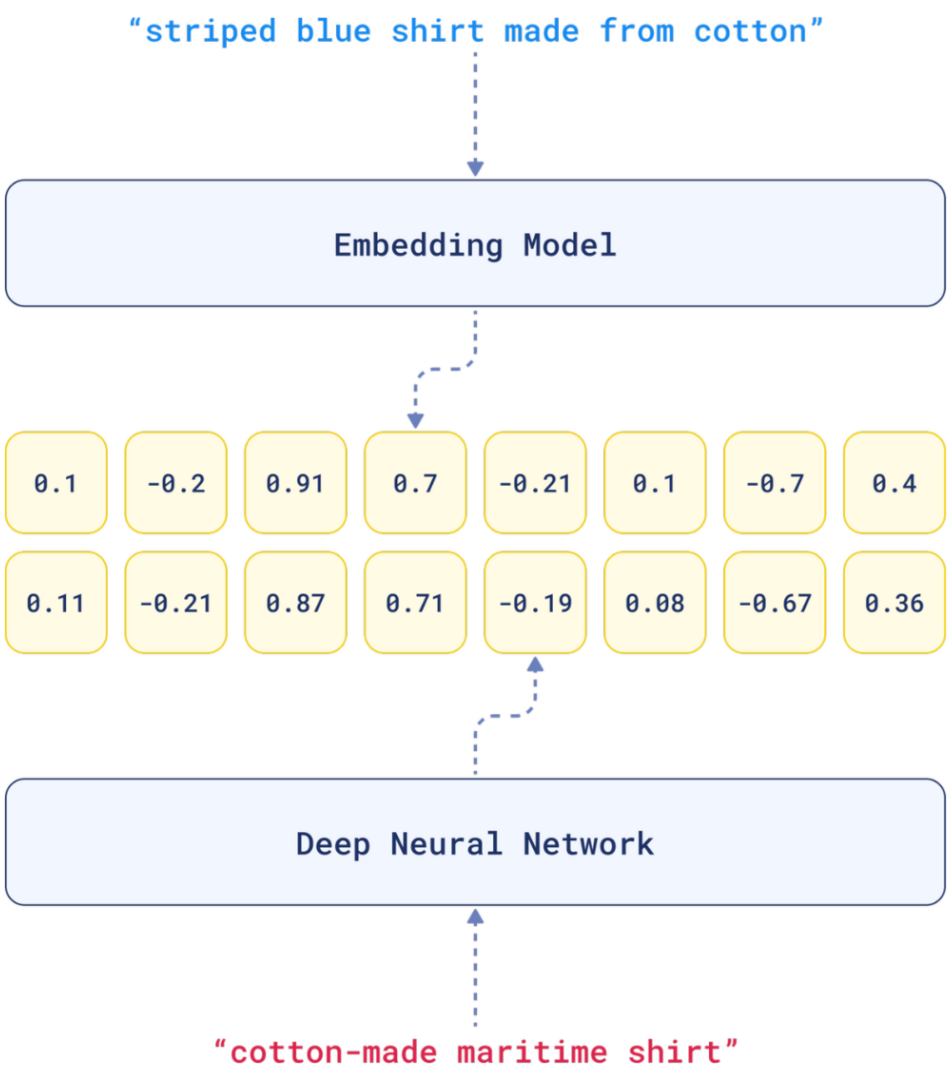
Indexing algorithms

Searching for the closest points in a plane: https://www.infoarena.ro/problema/cmap



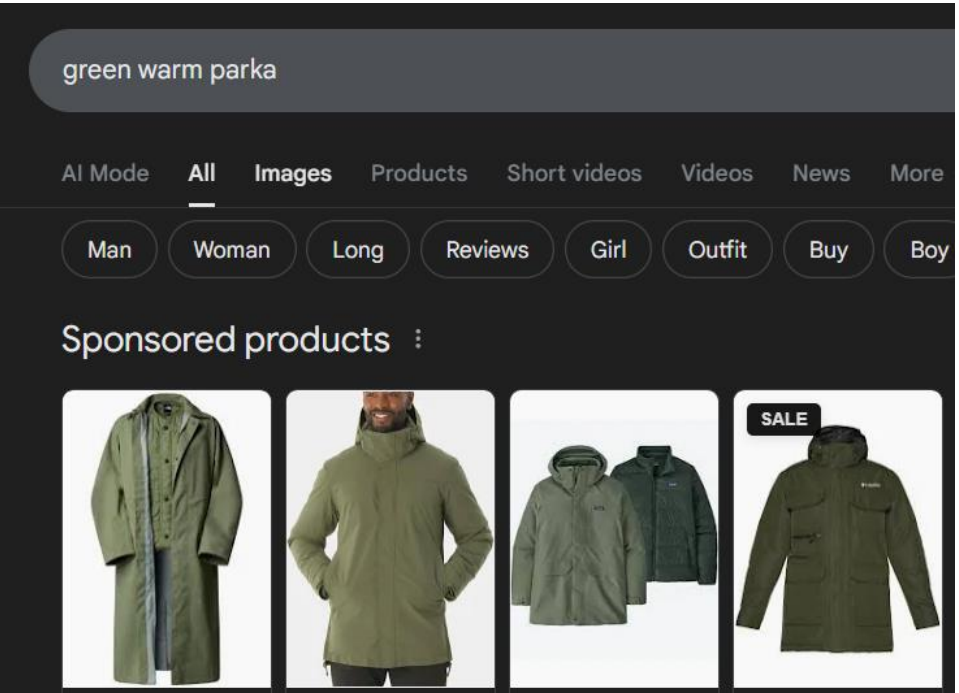
Features of a Vector DB

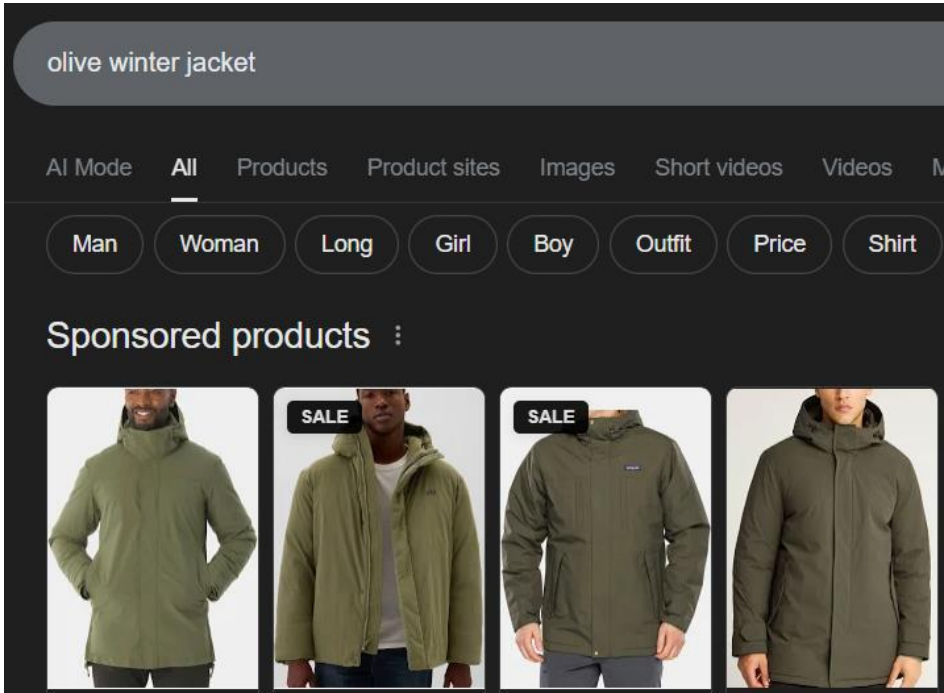
Similarity & Semantic Search



Features of a Vector DB

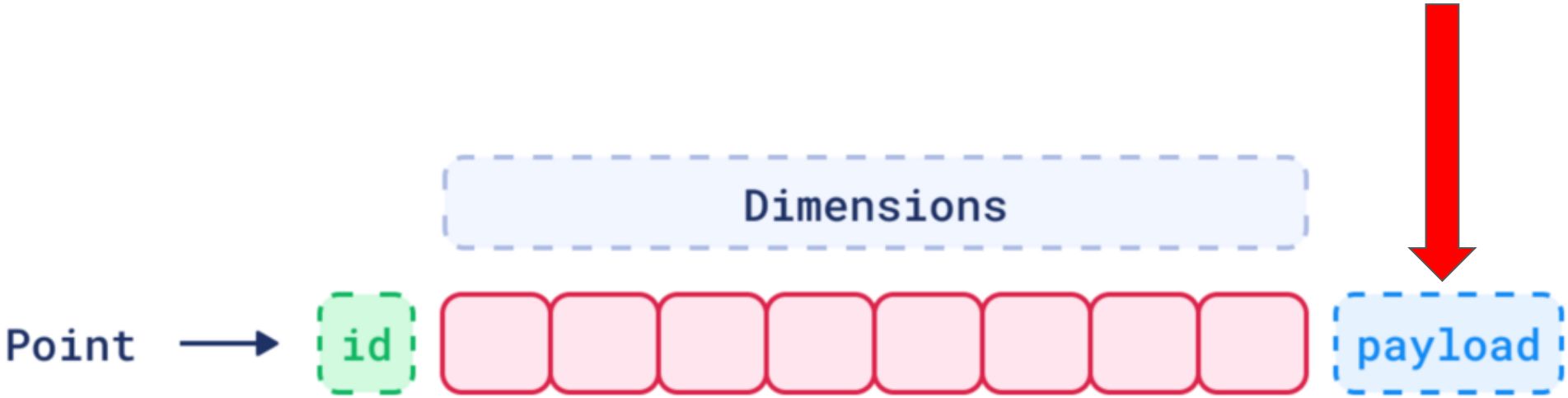
Similarity & Semantic Search





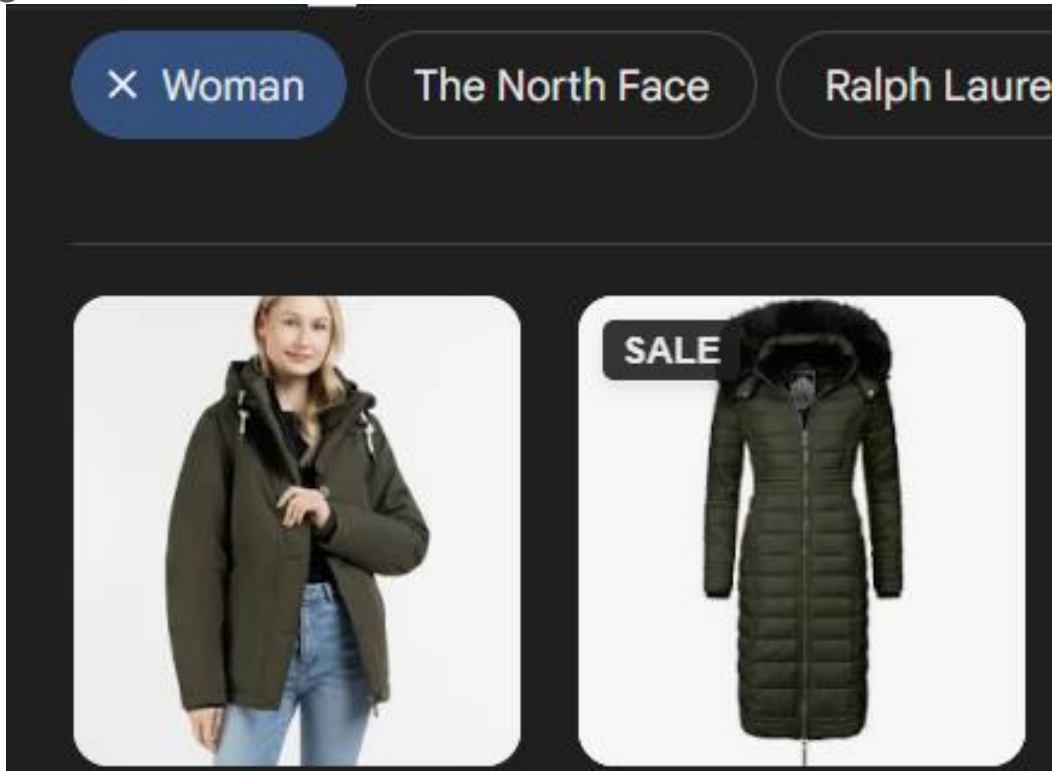
Data representation

Point = Id + Data itself + vector representation + **METADATA**



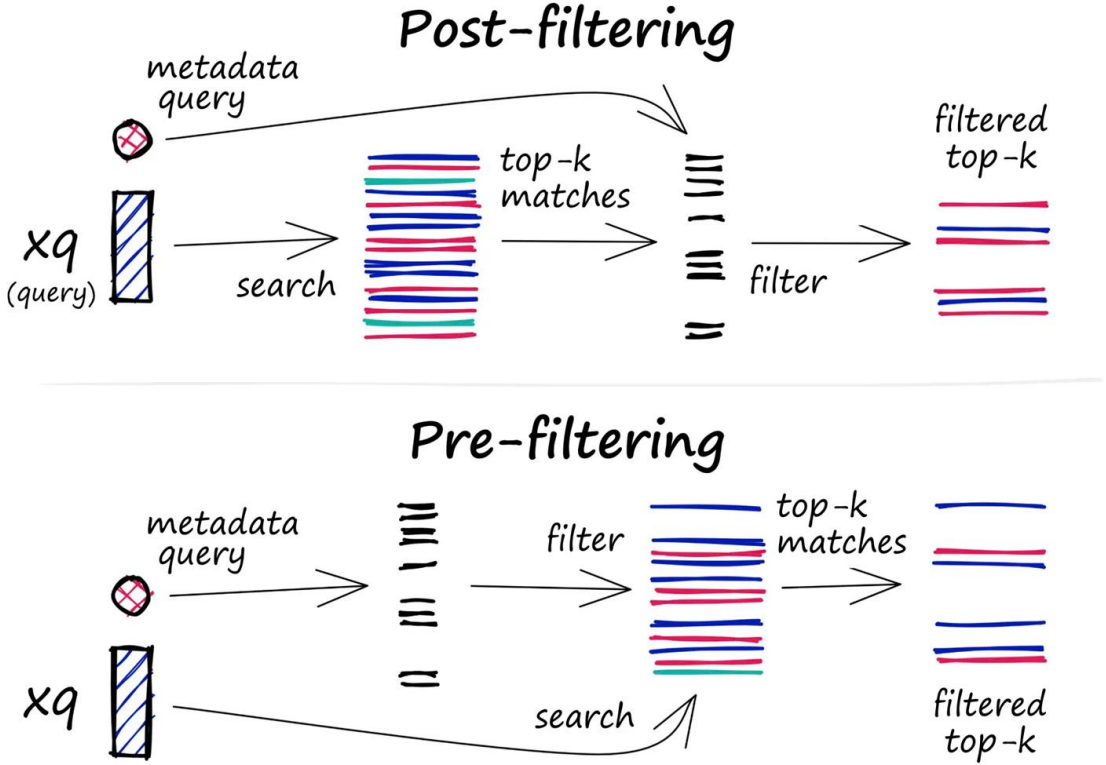
Features of a Vector DB

Metadata filtering



Features of a Vector DB

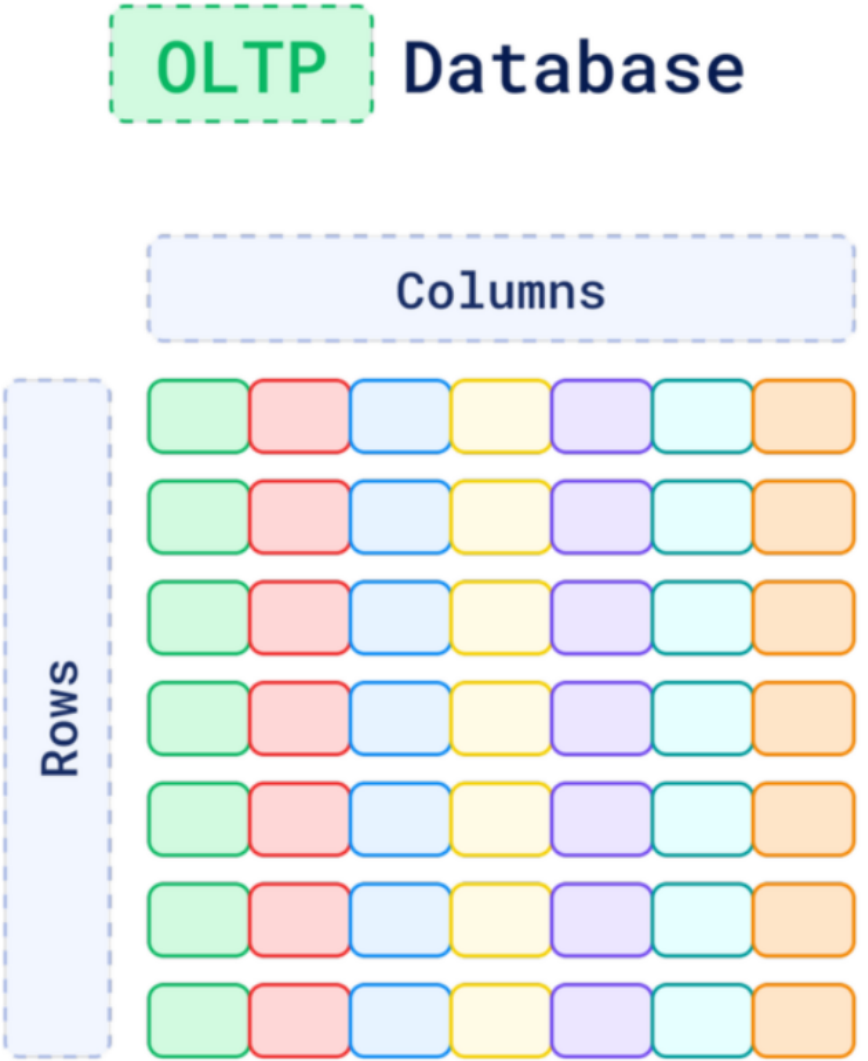
Metadata filtering



Relational DBs vs Vector DBs

Relational DBs:

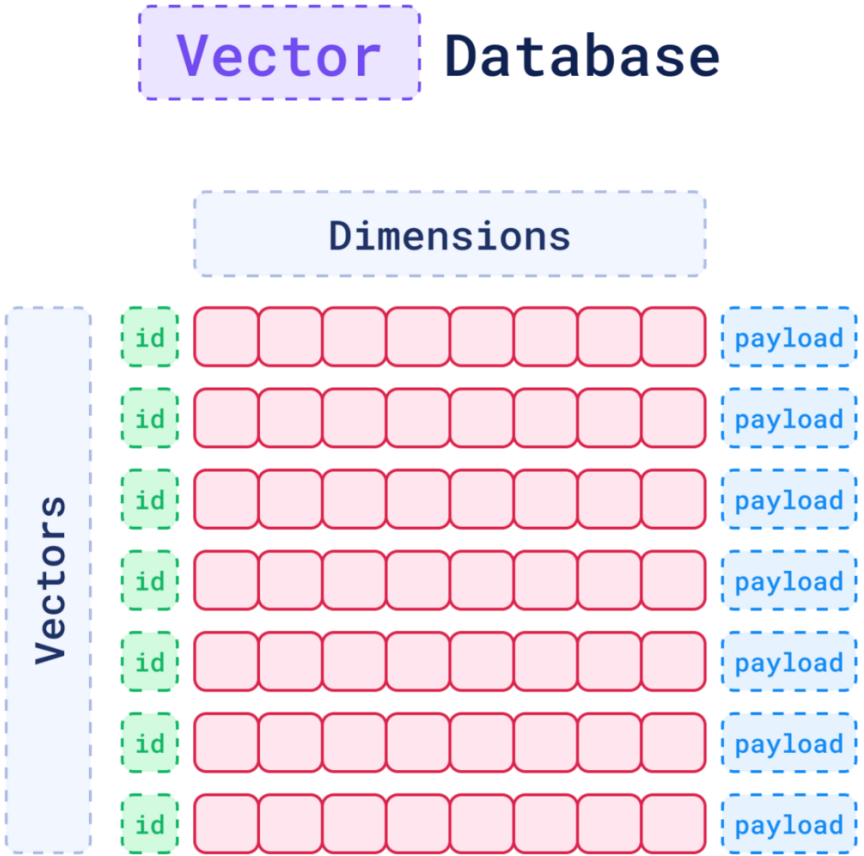
- ACID
- Storage and persistence of data
- Transactions Transaction Transactions
- Table joins, data aggregation
- Precise searches, exact matches
- ~~Horizontal scaling~~ Vertical scaling
- Standard data types



Relational DBs vs Vector DBs

Vector DBs:

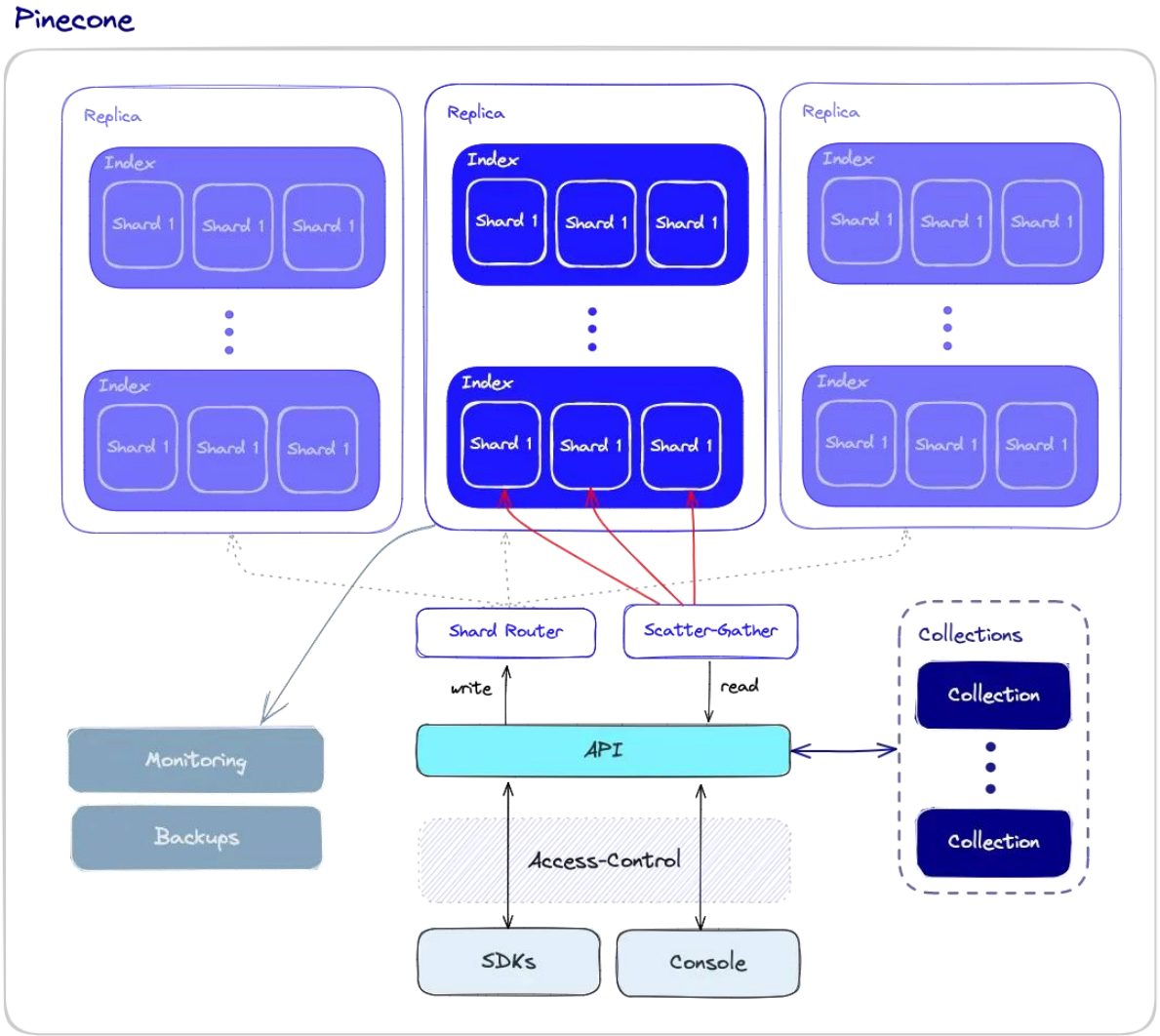
- Search Focused
- Optimized for approximate results
- If it can be vectorized, it can be searched
- Horizontal scaling + Vertical scaling



Types of Vector DBs

Fully fledged:

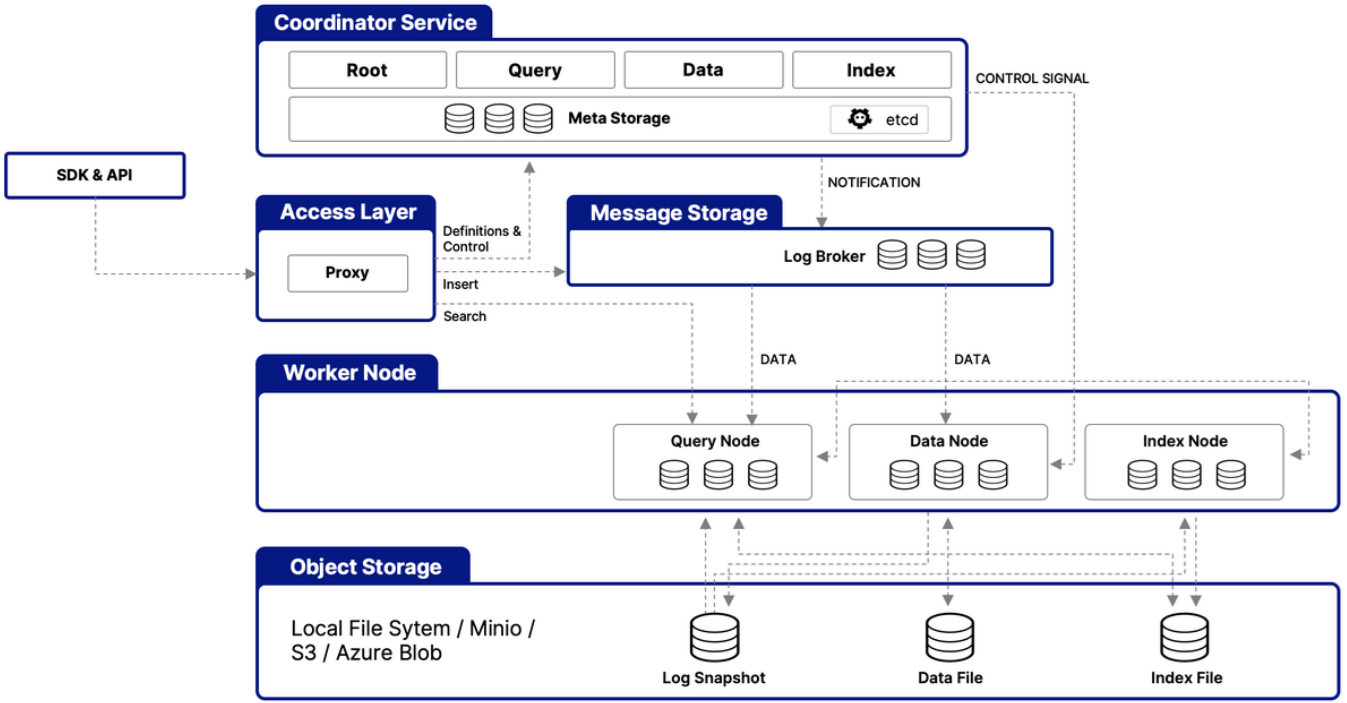
- Pinecone DB



Types of Vector DBs

Fully fledged:

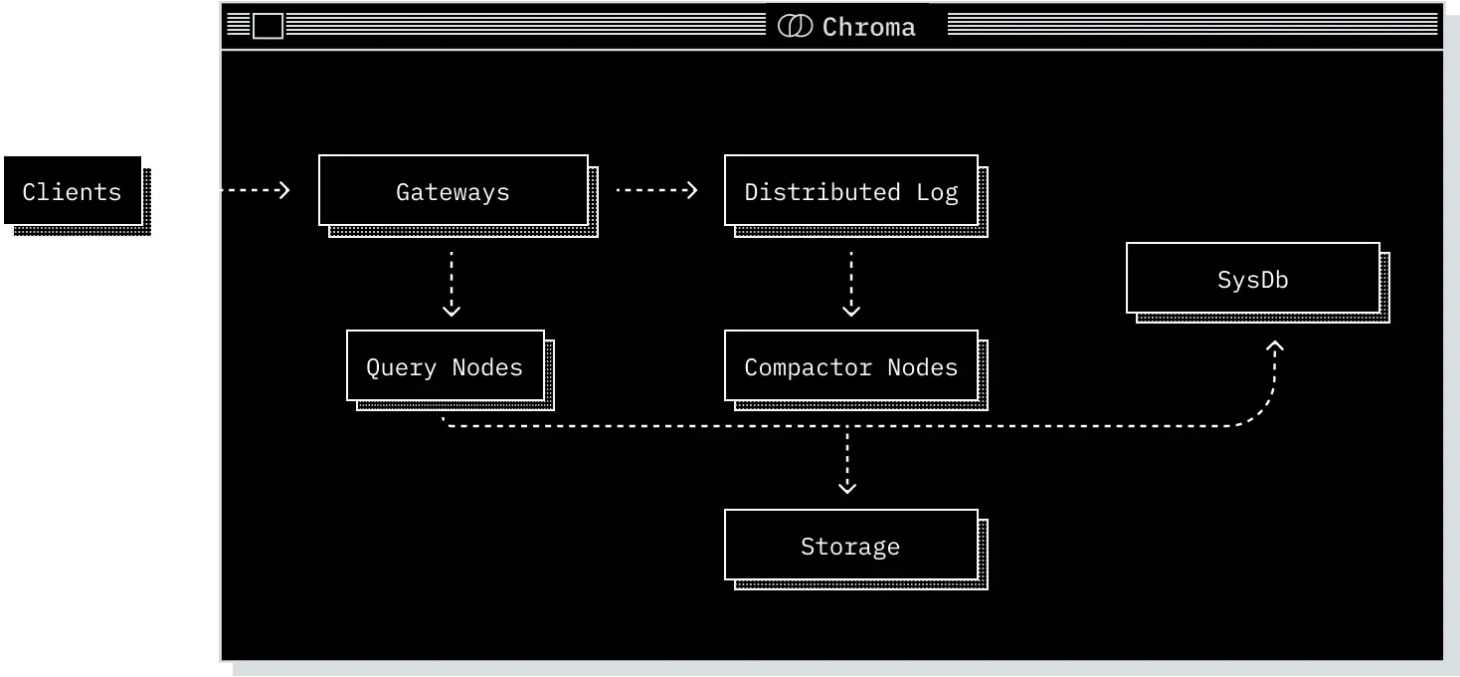
- Pinecone DB
- Milvus



Types of Vector DBs

Fully fledged:

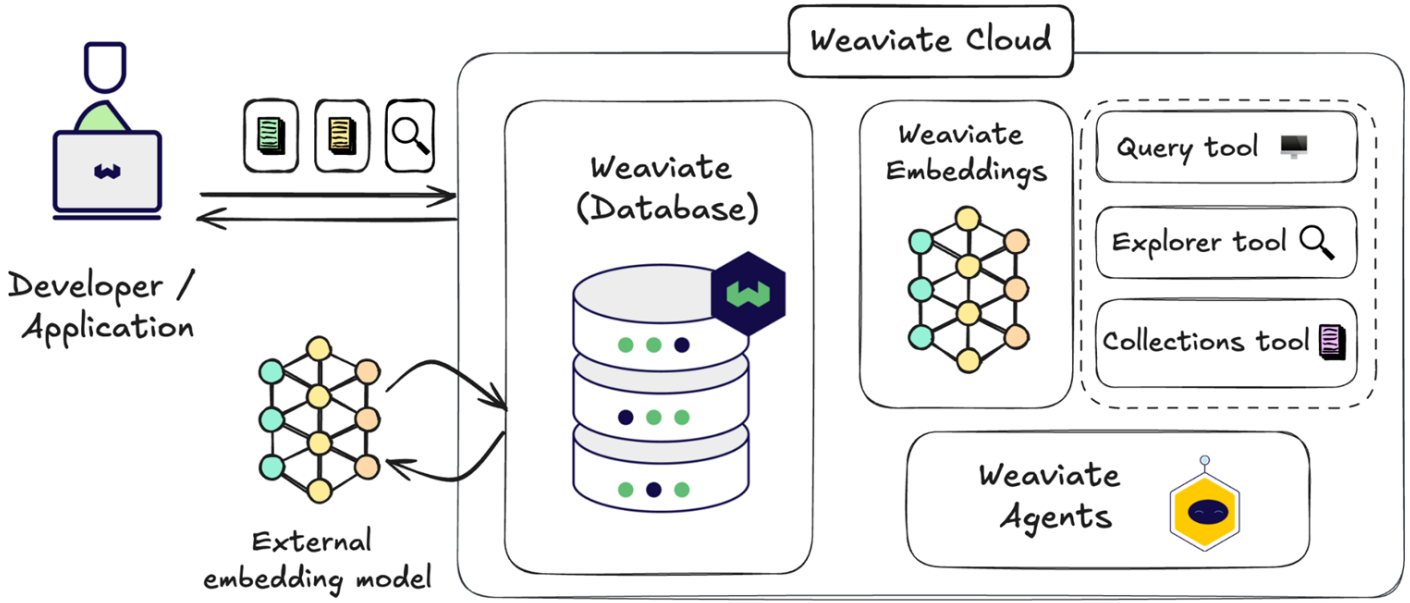
- Pinecone DB
- Milvus
- Chroma DB



Types of Vector DBs

Fully fledged:

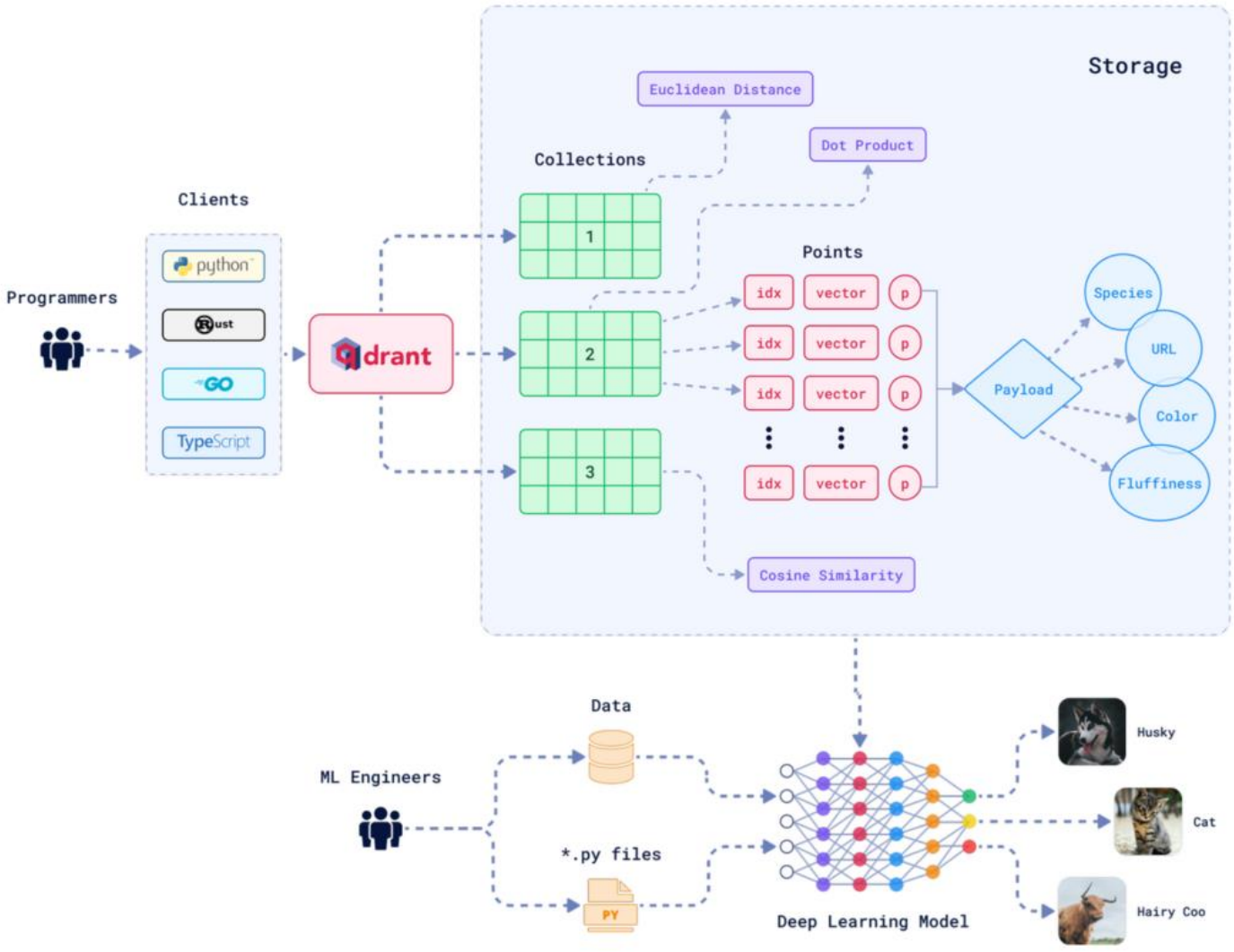
- Pinecone DB
- Milvus
- Chroma DB
- Weaviate



Types of Vecto

Fully fledged:

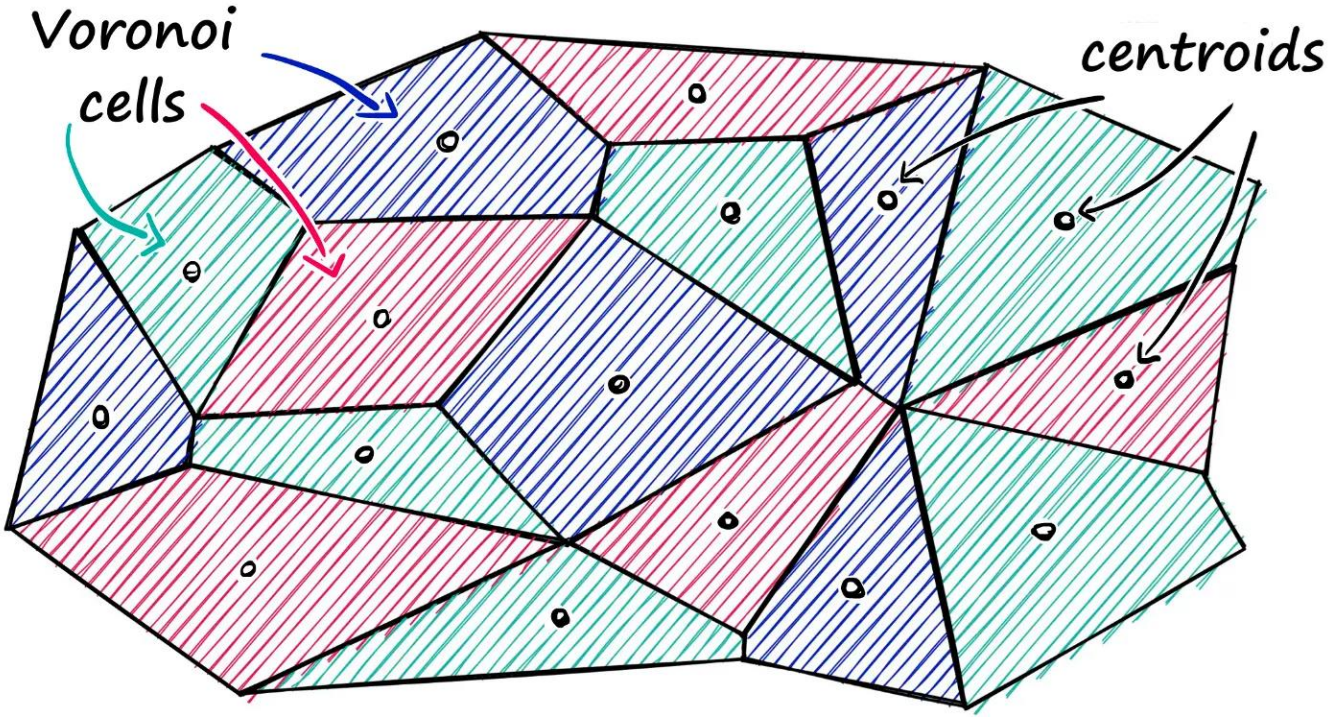
- Pinecone DB
- Milvus
- Chroma DB
- Weaviate
- Qdrant



Types of Vector DBs

Vector search library:

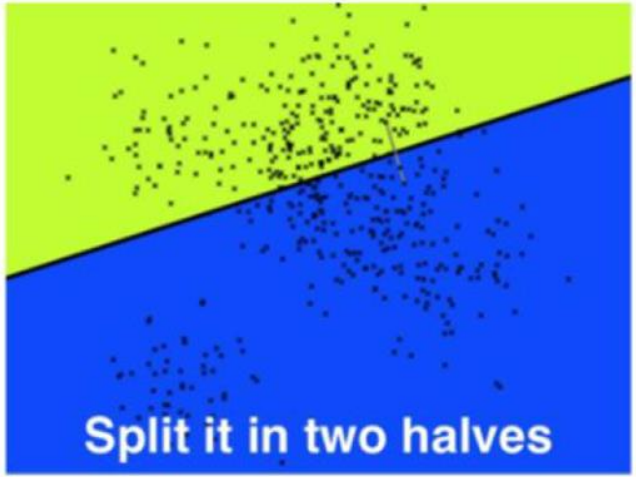
- FAISS

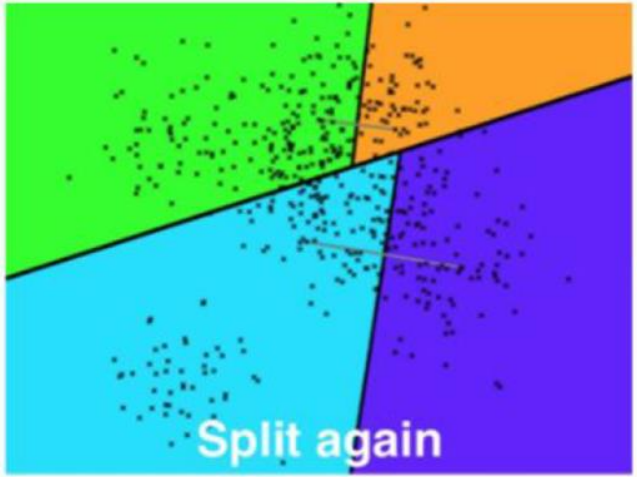


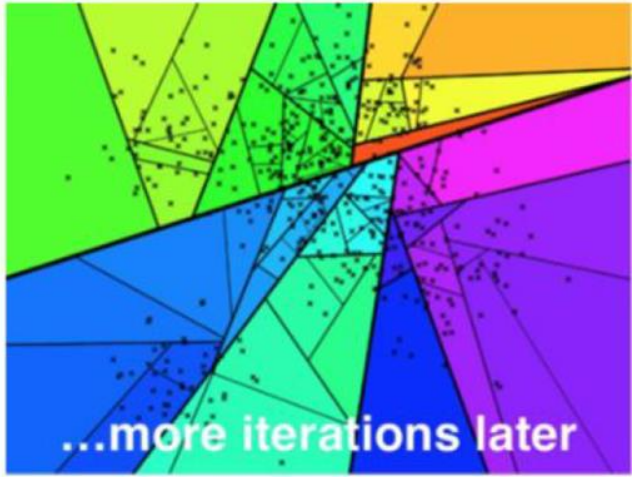
Types of Vector DBs

Vector search library:

- FAISS
- Annoy



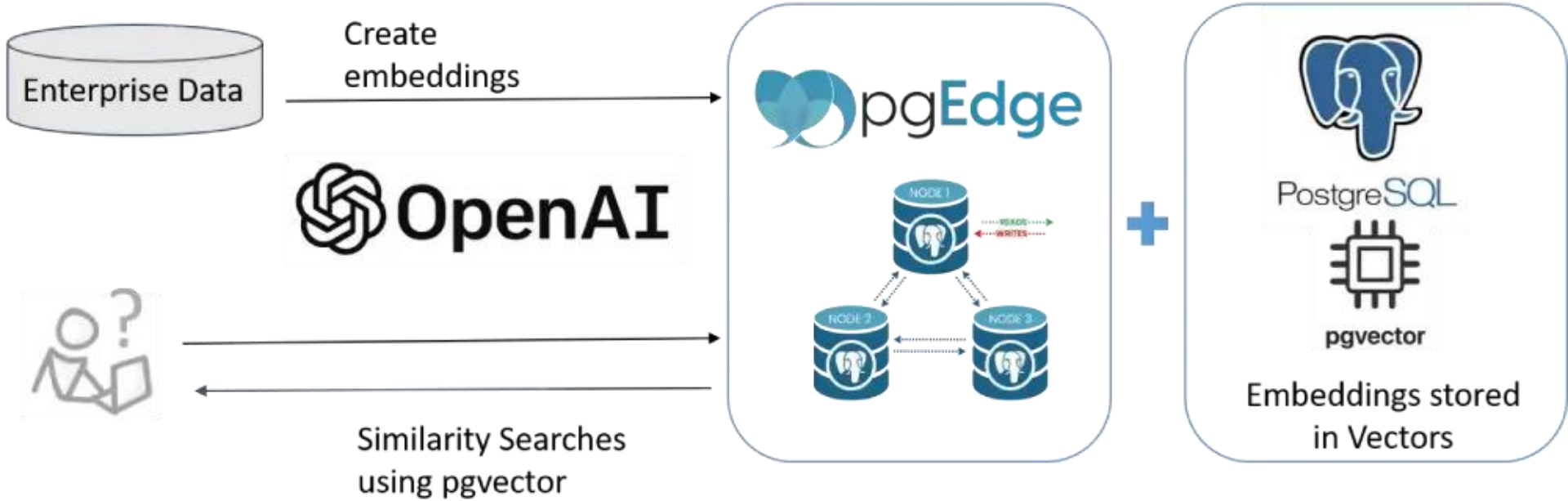




Types of Vector DBs

RDB Plugins:

- pgVector



References

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https://github.com/spotify/annoy

https://docs.weaviate.io/weaviate/concepts/data

https://developers.google.com/machine-learning/crash-course/embeddings